

Table 3.1 Summary of Important Equations

Quantity	Relationship	Values of Constants and Parameters (for Intrinsic Si at $T = 300$ K)
Carrier concentration in intrinsic silicon (cm^{-3})	$n_i = BT^{3/2}e^{-E_g/2kT}$	$B = 7.3 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-3/2}$ $E_g = 1.12 \text{ eV}$ $k = 8.62 \times 10^{-5} \text{ eV/K}$ $n_i = 1.5 \times 10^{10} / \text{cm}^3$
Diffusion current density (A/cm^2)	$J_p = -qD_p \frac{dp}{dx}$ $J_n = qD_n \frac{dn}{dx}$	$q = 1.60 \times 10^{-19} \text{ coulomb}$ $D_p = 12 \text{ cm}^2/\text{s}$ $D_n = 34 \text{ cm}^2/\text{s}$
Drift current density (A/cm^2)	$J_{\text{drift}} = q(p\mu_p + n\mu_n)E$	$\mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s}$ $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$
Resistivity ($\Omega \cdot \text{cm}$)	$\rho = 1/[q(p\mu_p + n\mu_n)]$	μ_p and μ_n decrease with the increase in doping concentration
Relationship between mobility and diffusivity	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$	$V_T = kT/q \approx 25.8 \text{ mV}$
Carrier concentration in n -type silicon (cm^{-3})	$n_{n0} \approx N_D$ $p_{n0} = n_i^2/N_D$	
Carrier concentration in p -type silicon (cm^{-3})	$p_{p0} \approx N_A$ $n_{p0} = n_i^2/N_A$	
Junction built-in voltage (V)	$V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$	
Width of depletion region (cm)	$\frac{x_n}{x_p} = \frac{N_A}{N_D}$ $W = x_n + x_p$ $= \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 + V_R)}$	$\epsilon_s = 11.7\epsilon_0$ $\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$
Charge stored in depletion layer (coulomb)	$Q_J = q \frac{N_A N_D}{N_A + N_D} A W$	
Forward current (A)	$I = I_p + I_n$ $I_p = Aq n_i^2 \frac{D_p}{L_p N_D} (e^{V/V_T} - 1)$ $I_n = Aq n_i^2 \frac{D_n}{L_n N_A} (e^{V/V_T} - 1)$	
Saturation current (A)	$I_S = Aq n_i^2 \left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right)$	
I - V Relationship	$I = I_S (e^{V/V_T} - 1)$	

Table 3.1 continued

Quantity	Relationship	Values of Constants and Parameters (for Intrinsic Si at $T = 300$ K)
Minority-carrier lifetime (s)	$\tau_p = L_p^2/D_p$ $\tau_n = L_n^2/D_n$	$L_p, L_n = 1 \mu\text{m to } 100 \mu\text{m}$ $\tau_p, \tau_n = 1 \text{ ns to } 10^4 \text{ ns}$
Minority-carrier charge storage (coulomb)	$Q_p = \tau_p I_p$ $Q_n = \tau_n I_n$ $Q = Q_p + Q_n = \tau_T I$	
Depletion capacitance (F)	$C_{j0} = A \sqrt{\left(\frac{\epsilon_s q}{2}\right) \left(\frac{N_A N_D}{N_A + N_D}\right) \frac{1}{V_0}}$ $C_j = C_{j0} \left(1 + \frac{V_R}{V_0}\right)^m$	$m = \frac{1}{3} \text{ to } \frac{1}{2}$
Diffusion capacitance (F)	$C_d = \left(\frac{\tau_T}{V_T}\right) I$	

PROBLEMS

Problems are marked with asterisks to describe their degree of difficulty. Difficult problems are marked with an asterisk (*); more difficult problems with two asterisks (**); and very challenging and/or time-consuming problems with three asterisks (***). Also, if in the following problems the need arises for the values of particular parameters or physical constants that are not stated, please consult Table 3.1.

Section 3.1: Intrinsic Semiconductors

3.1 Find values of the intrinsic carrier concentration n_i for silicon at -70°C , 0°C , 20°C , 100°C , and 125°C . At each temperature, what fraction of the atoms is ionized? Recall that a silicon crystal has approximately 5×10^{22} atoms/cm³.

3.2 Calculate the value of n_i for gallium arsenide (GaAs) at $T = 300$ K. The constant $B = 3.56 \times 10^{14}$ (cm⁻³K^{-3/2}) and the bandgap voltage $E_g = 1.42$ eV.

Section 3.2: Doped Semiconductors

3.3 For a p -type silicon in which the dopant concentration $N_A = 10^{18}/\text{cm}^3$, find the hole and electron concentrations at $T = 300$ K.

3.4 For a silicon crystal doped with phosphorus, what must N_D be if at $T = 300$ K the hole concentration drops below the intrinsic level by a factor of 10^7 ?

3.5 In a phosphorus-doped silicon layer with impurity concentration of $10^{16}/\text{cm}^3$, find the hole and electron concentrations at 27°C and 125°C .

Section 3.3: Current Flow in Semiconductors

3.6 A young designer, aiming to develop intuition concerning conducting paths within an integrated circuit, examines the end-to-end resistance of a connecting bar $10 \mu\text{m}$ long, $3 \mu\text{m}$ wide, and $1 \mu\text{m}$ thick, made of various materials. The designer considers: